

Chandan Abhishek Muchukota

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Summary

Software Engineer with experience building distributed backend systems, cloud-native APIs, and **AI-powered** applications. Proven track record shipping production services at scale, with hands-on work in **large language models**, **retrieval-augmented generation**, and conversational interfaces integrated into enterprise workflows.

Accomplishments

- Served as a **founding engineer** at multiple early-stage startups (CodeGreenBack, Phamoss, CISApp), leading development of scalable web and mobile platforms.
- Engineered the **fastest online compiler** for the CodeGreenBack platform using **Go (Golang)** and developed large-scale backend systems including **production APIs serving 1M+ users** and **distributed pipelines processing 10M+ events/day**.
- Solved **500+ LeetCode problems**, achieving a 3-star rating and demonstrating strong algorithmic problem-solving ability.

Experience

University of Florida

Jan 2025 - Present

Software Engineer

Gainesville, FL

- Architected a geospatial digital-twin platform for resilience planning using **React, TypeScript, ArcGIS JS SDK, Dash, and Plotly**, leveraging a microservices architecture to enable interactive analysis across 250K+ geospatial building records.
- Designed robust geospatial APIs leveraging **FastAPI, SQLAlchemy, and PostGIS** and created comprehensive documentation and onboarding materials, reducing new developer ramp-up time by **40%**.
- Constructed spatial analytics pipelines processing **250K+ building records** and **1M+ metadata attributes**, uncovering flood, heat, and infrastructure vulnerabilities that informed urban planning decisions and scenario-based resilience simulations.
- Orchestrated ETL pipelines using **Apache Airflow on GCP**, processing **10+ GB geospatial datasets** and loading curated data into BigQuery improving analytical query performance by **30%**.
- Designed and developed a **multi-agent conversational assistant** using **Gemini APIs**, where specialized agents handled distinct tasks including geospatial queries, resilience scenario simulations, and analytical workflow execution, coordinated through an **LLM-based orchestration layer** with **tool-calling** capabilities to interface with backend geospatial services.

PruTech (Client: NYC Comptroller's Office)

May 2025 - Dec 2025

Software Engineer Intern

Remote, USA

- Engineered secure REST APIs using **C#, .NET Core**, and SQL Server supporting financial systems processing \$2B+ annually, and implemented **logging and monitoring using Azure Monitor and Application Insights** to ensure reliability and compliance in production environments.
- Optimized high-volume **SQL Server** workloads through indexing and query tuning improving performance by **40%**.
- Built interactive dashboards using **Angular** integrated with **.NET APIs** reducing report generation time by **25%**.
- Automated testing and CI/CD pipelines across the **software development lifecycle (SDLC)** using **NUnit and Azure DevOps**, collaborating with cross-functional teams to reduce production defects by **25%**.
- Migrated and secured **2TB+ financial datasets** ensuring compliance with enterprise security standards.

Infinite Analytics

Dec 2023 - Jul 2024

Software Engineer

Mumbai, India

- Built and maintained high-throughput backend services on AWS using FastAPI, powering production recommendation APIs for over 1M users with sub-200ms latency and 99.99% uptime, and integrated **centralized logging and monitoring using Splunk** to analyze system behavior and debug production issues.
- Integrated LLM-powered code review automation into CI/CD workflows, leveraging natural language processing to generate actionable suggestions and improve development productivity by **30%**.
- Designed event-driven ingestion pipelines using Kafka and RabbitMQ, implementing **Retrieval-Augmented Generation (RAG)** to enrich real-time data streams and reduce processing latency to under 5 seconds.

Cogoport

Feb 2023 - Oct 2023

Software Engineer

Mumbai, India

- Built backend microservices using FastAPI and Ruby on Rails on AWS, powering recommendation, forecasting, and analytics APIs serving 1M+ monthly users, and implemented **monitoring and observability using Grafana dashboards** to track service health, latency, and system performance.
- Developed an internal **LLM-powered search engine** for the logistics sales team to query large volumes of trade rules, regulations, and shipment documentation using **Retrieval-Augmented Generation (RAG)**, significantly reducing manual lookup time.
- Extended the search engine into a conversational **booking bot** that enabled sales agents to initiate and complete shipment bookings through natural language, integrating with backend logistics APIs.
- Designed a custom **demand forecasting system** for container shipping logistics, incorporating seasonality and trend modeling to achieve 78% prediction accuracy.
- Reduced excess inventory by **20%** and shipment cancellations by **37%** improving logistics planning and capacity utilization.

Projects

Delphi Pascal Interpreter

- Implemented a custom Delphi Pascal interpreter in **Java** using **ANTLR** supporting object-oriented language features.
- Designed parsing, AST traversal, and execution engine using the Visitor pattern and Java collections to efficiently implement language semantics
- Improved interpreter performance by 25% through compile-time evaluation, memory pool optimization, and profiling with Java performance tools

Technical Skills

- Languages:** Python, C#, Java, JavaScript, Go, SQL
- Frameworks:** FastAPI, Ruby on Rails, .NET Core, React, Angular, Node.js, Flutter
- Cloud / Infrastructure:** AWS, Azure, Docker, Apache Airflow, BigQuery
- Databases:** PostgreSQL, MySQL, Cassandra, SQL Server, PostGIS
- Messaging / Streaming:** Kafka, RabbitMQ
- Operating Systems:** Linux, Unix
- Concepts:** Distributed Systems, Microservices, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), REST APIs, System Design, Data Structures & Algorithms, CI/CD, Scalability, Performance Optimization

Education

University of Florida | *Master of Science, Computer Science* (GPA: 3.95)

Gainesville, FL Aug 2024 - May 2026